CS413 — Problem Solving and Decision Making

50 Multiple-Choice Questions

Lectures 1 & 2

Practice set with answers and explanations

All questions are written in English and based on the uploaded lecture slides.
Each answer includes a short explanation and the exact PDF page / slide reference.

Prepared for revision and self-assessment

Answer key is shown directly below each question.

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Quick Answer Key 17

Multiple-Choice Questions

Question 1

Which statement best defines a Problem Statement?

A. A detailed list of project tasks

B. A clear description of an issue, challenge, or gap that identifies the problem and the need for action

C. A proposed budget for a project

D. A final report describing the solution

Correct answer: B
Why? It clearly describes the issue or gap and defines the need for an effective solution and action.
Source: PDF p. 1 — Slide 2

Question 2

A good Problem Statement should explain:

A. Only the proposed solution

B. The issue and why it needs attention

C. The project schedule only

D. The personal opinions of the team

Correct answer: B
Why? A good statement clearly explains the issue and why it needs attention.
Source: PDF p. 2 — Slide 3

Question 3

Which qualities should a good Problem Statement have?

A. Complex, broad, and subjective

B. Simple, specific, and objective

C. Long, technical, and emotional

D. Vague, general, and solution-focused

Correct answer: B
Why? The slide says the challenge should be defined in a simple, specific, and objective way.
Source: PDF p. 2 — Slide 3

Question 4

Which of the following is NOT one of the five Ws listed for an effective Problem Statement?

A. Who is affected?

B. Where does it occur?

C. How much will the solution cost?

D. Why is solving it important?

Correct answer: C
Why? The five questions are Who, What, Where, When, and Why; cost is not one of them.
Source: PDF p. 2 — Slide 3

Question 5

Why are Problem Statements important for clarifying focus?

A. They encourage teams to treat symptoms only

B. They prevent teams from understanding the issue

C. They help teams understand the issue and avoid tackling symptoms instead of root causes

D. They replace the need for project goals

Correct answer: C
Why? Clarifying focus means understanding the issue deeply and avoiding symptom-only treatment.
Source: PDF p. 3 — Slide 4

Question 6

How can a Problem Statement enhance communication?

A. By hiding the issue from stakeholders

B. By aligning stakeholders, team members, and sponsors on what is being solved and why

C. By reducing all communication to one person

D. By removing the need for documentation

Correct answer: B
Why? A clear statement aligns the involved people on the problem and its importance.
Source: PDF p. 3 — Slide 4

Question 7

Which sequence matches the key steps for writing a Problem Statement?

A. Find the solution → identify the team → estimate cost → launch

B. Identify the problem → put it into context → find the root cause → describe the ideal outcome → propose a solution and outline benefits

C. Plan the budget → hire staff → test → close

D. Describe benefits → select tools → identify symptoms → stop

Correct answer: B
Why? This is the five-step sequence shown in the slide diagram.
Source: PDF p. 4 — Slide 5

Question 8

Problem solving depends on correctly identifying the problem's:

A. Color

B. Cause

C. Owner's job title

D. Presentation format

Correct answer: B
Why? The slide says the cause must be correctly identified so the best solution can be selected and implemented for sustained results.
Source: PDF p. 6 — Slide 7

Question 9

Which sequence represents the problem-solving cycle shown in the diagram?

A. Define the problem → diagnose the root cause → identify and implement a solution → sustain the results

B. Implement a solution → define the problem → stop

C. Diagnose the cause → ignore results → restart

D. Sustain results → select a problem → define a budget

Correct answer: A
Why? The diagram presents these four connected stages in this order.
Source: PDF p. 6 — Slide 7

Question 10

When defining a problem, why should fact be differentiated from opinion?

A. To focus on the problem rather than its symptoms

B. To make the statement longer

C. To avoid collecting evidence

D. To replace subject matter experts

Correct answer: A
Why? The slide emphasizes stating the problem specifically, separating fact from opinion, and discovering underlying causes.
Source: PDF p. 7 — Slide 8

Question 11

During problem definition, whom should the team involve?

A. Only the project manager

B. All interested workgroups affected by the problem and subject matter experts

C. Only external customers

D. No one outside the development team

Correct answer: B
Why? The slide recommends involving affected workgroups and asking subject matter experts to understand processes and issues.
Source: PDF p. 7 — Slide 8

Question 12

Which type of supporting data should be collected and analyzed when defining a problem?

A. Qualitative and quantitative

B. Only financial

C. Only opinions

D. Only historical photographs

Correct answer: A
Why? The slide explicitly calls for collecting and analyzing both qualitative and quantitative supporting data.
Source: PDF p. 7 — Slide 8

Question 13

A gemba review is used to:

A. Document how processes currently work

B. Choose a project name

C. Estimate future market share

D. Replace all data collection

Correct answer: A
Why? The slide says to go to the gemba review and document how processes currently work, including who does what and when.
Source: PDF p. 7 — Slide 8

Question 14

In the descriptive questions, “Where” refers to:

A. The person who detected the issue

B. The point of detection where the problem was detected

C. The reason the issue matters

D. The number of affected products

Correct answer: B
Why? The slide defines Where as the point of detection.
Source: PDF p. 8 — Slide 9

Question 15

In the descriptive questions, "Who" refers to:

A. Who caused the problem

B. Who detected the problem

C. Who will pay for the solution

D. Who approves the budget

Correct answer: B
Why? The slide explicitly says Who is for detection only, not cause.
Source: PDF p. 8 — Slide 9

Question 16

Root Cause Analysis (RCA) is a collective term for:

A. Approaches, tools, and techniques used to uncover causes of problems and direct corrective action

B. A method for writing project advertisements

C. A process for assigning grades

D. A technique for estimating team size

Correct answer: A
Why? This is the definition of RCA given in the lecture.
Source: PDF p. 9 — Slide 10

Question 17

Which question is associated with the 5 Whys technique?

A. Why? Why? Why? Why? Why?

B. Who? Where? When? How? What?

C. How much? How many? How long?

D. Which tool? Which team? Which budget?

Correct answer: A
Why? The slide identifies repeated Why questions as the 5 Whys technique.
Source: PDF p. 9 — Slide 10

Question 18

When generating potential solutions, the team should:

A. Select the first idea immediately

B. Postpone selecting one solution until several alternatives have been proposed

C. Avoid short- and long-term alternatives

D. Ignore customers and suppliers

Correct answer: B
Why? The slide recommends generating several alternatives before choosing one.
Source: PDF p. 10 — Slide 11

Question 19

Why should multiple alternatives be considered?

A. They can significantly improve the value of the ideal solution

B. They always eliminate implementation costs

C. They make root-cause analysis unnecessary

D. They guarantee that no risks exist

Correct answer: A
Why? The lecture explains that considering alternatives can enhance the value of the ideal solution.
Source: PDF p. 10 — Slide 11

Question 20

Which action is suggested to encourage better solutions?

A. Combine acceptable ideas into a stronger hybrid solution

B. Stop brainstorming after the first idea

C. Exclude customers and suppliers

D. Avoid creative thinking

Correct answer: A
Why? The slide suggests combining two acceptable solutions into one stronger hybrid solution.
Source: PDF p. 10 — Slide 11

Question 21

Which of the following is a question to consider when evaluating a solution?

A. How does this solution address the root cause?

B. What color should the report be?

C. Who created the lecture slides?

D. How can monitoring be avoided?

Correct answer: A
Why? The slide asks whether the proposed solution addresses the root cause.
Source: PDF p. 11 — Slide 12

Question 22

A contingency plan is concerned with:

A. Possible problems with the solution and how to respond

B. The original problem title only

C. The team's attendance record

D. Removing all monitoring activities

Correct answer: A
Why? The slide asks about possible problems with the solution and contingency plans.
Source: PDF p. 11 — Slide 12

Question 23

After selecting a solution, implementation becomes easier with:

A. A carefully developed plan specifying what, who, and when

B. An undocumented verbal agreement

C. No measurement criteria

D. A plan without responsibilities

Correct answer: A
Why? The slide says the implementation plan should include details on what, who, and when.
Source: PDF p. 12 — Slide 14

Question 24

A carefully developed implementation plan should determine how the solution will be:

A. Measured

B. Renamed

C. Advertised

D. Archived before use

Correct answer: A
Why? The plan should define how the change will be measured and how implementation will be monitored.
Source: PDF p. 12 — Slide 14

Question 25

Which is an example of a potential implementation barrier?

A. Human factors that may make people feel threatened by new procedures

B. The existence of a clear goal

C. A documented measurement method

D. A defined project deliverable

Correct answer: A
Why? The slide highlights human factors and the difficulty of remembering new procedures as considerations.
Source: PDF p. 12 — Slide 14

Question 26

Planning is defined in the lecture as:

A. Organizing pieces of knowledge into a consistent sequence of actions to accomplish a goal

B. Randomly selecting actions

C. Recording completed work only

D. Choosing a goal without actions

Correct answer: A
Why? This is the definition of planning shown in the lecture.
Source: PDF p. 13 — Slide 16

Question 27

The task of a planner is to find:

A. A sequence of actions that allows a problem solver to accomplish a specific task

B. A list of unrelated facts

C. The cheapest robot available

D. A final state without actions

Correct answer: A
Why? The slide explains that a planner finds an action sequence to accomplish a specific task.
Source: PDF p. 13 — Slide 16

Question 28

Traditional planning is described as knowledge-intensive because:

A. Plan creation requires organizing pieces of knowledge and partial plans into a solution procedure

B. It does not use knowledge

C. It relies only on random choices

D. It avoids solution procedures

Correct answer: A
Why? The lecture directly links traditional planning's knowledge-intensive nature to organizing knowledge and partial plans.
Source: PDF p. 13 — Slide 16

Question 29

Which is an application of planning mentioned in the lecture?

A. Manufacturing process control

B. Social media filtering only

C. Image compression only

D. Audio recording only

Correct answer: A
Why? The slide mentions manufacturing applications such as process control and designing a robot controller.
Source: PDF p. 13 — Slide 16

Question 30

In the block-planning example, what is the first listed action?

A. Go to room b

B. Put down whatever is now held

C. Pick up block a

D. Return to the original location

Correct answer: B
Why? The listed sequence starts by putting down whatever is currently held.
Source: PDF p. 14 — Slide 17

Question 31

The actions of a traditional robot are composed of the robot's:

A. Atomic actions

B. Opinions

C. Goals only

D. Visual labels

Correct answer: A
Why? The lecture states that the steps of a traditional robot are composed of atomic actions.
Source: PDF p. 15 — Slide 18

Question 32

Plans are created by searching through:

A. A space of possible actions

B. A list of unrelated objects

C. A fixed set of opinions

D. Only the final goal state

Correct answer: A
Why? The slide says plans are created by searching possible actions until the required sequence is discovered.
Source: PDF p. 15 — Slide 18

Question 33

The search for a plan terminates when:

A. The goal state is produced

B. The first action is considered

C. The robot becomes empty

D. All possible actions are removed

Correct answer: A
Why? The lecture says search terminates when the goal state, the appropriate description of the world, is produced.
Source: PDF p. 15 — Slide 18

Question 34

Traditional planning may be viewed as:

A. A state-space search

B. A database normalization process

C. A sorting algorithm

D. A communication protocol

Correct answer: A
Why? The slide explicitly says planning may be seen as a state-space search.
Source: PDF p. 16 — Slide 19

Question 35

In the blocks-world example, the robot arm can move a clear block to:

A. Any location on the tabletop or on top of another clear block

B. Only the original location

C. Only the floor

D. Any block even if covered

Correct answer: A
Why? The lecture says the gripper can grasp any clear block and move it to any tabletop location or onto another clear block.
Source: PDF p. 18 — Slide 20

Question 36

Which task means picking up block W from its current location?

A. putdown(W)

B. pickup(W)

C. stack(U,V)

D. unstack(U,V)

Correct answer: B
Why? The task table defines pickup(W) as picking up block W from its current location.
Source: PDF p. 19 — Slide 21

Question 37

Which operator places block U on top of block V?

A. pickup(U)

B. putdown(U)

C. stack(U,V)

D. unstack(U,V)

Correct answer: C
Why? The task table defines stack(U,V) as placing U on top of V.
Source: PDF p. 19 — Slide 21

Question 38

What does clear(X) mean in the state description?

A. Block X is on the table

B. Block X has nothing on top of it

C. The robot is holding block X

D. Block X is at coordinates X, Y, Z

Correct answer: B
Why? The predicate list defines clear(X) as having nothing on top of block X.
Source: PDF p. 20 — Slide 22

Question 39

What does gripping() represent?

A. The robot arm is holding block X

B. The robot gripper is empty

C. Block X is on the table

D. Block X is clear

Correct answer: B
Why? The slide defines gripping() as the robot gripper being empty; gripping(X) means it is holding block X.
Source: PDF p. 20 — Slide 22

Question 40

What does STRIPS stand for in the lecture?

A. Stanford Research Institute Planning System

B. Structured Robot Intelligence Processing Software

C. State Transition and Robot Instruction Planning System

D. Standard Research Information Processing System

Correct answer: A
Why? The slide expands STRIPS as Stanford Research Institute Planning System.
Source: PDF p. 21 — Slide 23

Question 41

In a STRIPS operator description, the first element is the:

A. Add list

B. Delete list

C. Preconditions list

D. Goal list

Correct answer: C
Why? The first element is the preconditions (P), which must be satisfied before applying an operator.
Source: PDF p. 22 — Slide 24

Question 42

In a STRIPS operator description, the add list contains:

A. Items removed from the state

B. Additions to the state description resulting from the operator

C. Conditions required before the operator

D. Actions that must never be used

Correct answer: B
Why? The add list (A) contains additions to the state description after applying the operator.
Source: PDF p. 22 — Slide 24

Question 43

In a STRIPS operator description, the delete list contains:

A. Items removed from the state description

B. New items added to the state

C. The operator's name only

D. The final goal only

Correct answer: A
Why? The delete list (D) contains items removed from the state description when the operator is applied.
Source: PDF p. 22 — Slide 24

Question 44

Which set correctly lists the four blocks-world operators?

A. move, rotate, stop, start

B. pickup, putdown, stack, unstack

C. define, diagnose, monitor, learn

D. push, pull, open, close

Correct answer: B
Why? The lecture identifies pickup, putdown, stack, and unstack as the four operators.
Source: PDF p. 21 and p. 23 — Slides 23 and 25

Question 45

The decision-making process is:

A. A systematic approach to choosing a course of action among multiple alternatives

B. A random response to a situation

C. A process that avoids gathering information

D. The same as implementing a solution without evaluation

Correct answer: A
Why? The slide defines decision-making as a systematic approach involving alternatives, information, and evaluation.
Source: PDF p. 24 — Slide 26

Question 46

Which criterion is mentioned for evaluating potential solutions in decision-making?

A. Feasibility, risk, and impact

B. Font size, color, and layout

C. Popularity only

D. Team seniority only

Correct answer: A
Why? The lecture lists feasibility, risk, and impact as criteria for evaluating potential solutions.
Source: PDF p. 24 — Slide 26

Question 47

Which of the following appears in the lecture's list of problem-solving skills?

A. Adaptability

B. Database indexing

C. Cryptography

D. Public speaking only

Correct answer: A
Why? Adaptability is included in the list of key problem-solving skills.
Source: PDF p. 26 — Slide 28

Question 48

Which of the following is listed as a decision-making skill?

A. Risk Assessment

B. Image Processing

C. Code Refactoring

D. Network Routing

Correct answer: A
Why? Risk Assessment appears in the lecture's decision-making skills list.
Source: PDF p. 25 — Slide 27

Question 49

Which is the correct order of the decision-making process presented in the slides?

A. Gather information → take action → identify problem → review

B. Identify the problem → gather relevant information → identify alternatives → weigh evidence → choose → take action → review

C. Choose an alternative → identify the problem → gather information → stop

D. Identify alternatives → take action → weigh evidence → define the problem

Correct answer: B
Why? This sequence follows all seven steps listed in the lecture.
Source: PDF p. 27 — Slide 30

Question 50

Which sequence matches the Problem-Solving Process slide?

A. Define → Analyze → Generate possible solutions → Evaluate and select → Implement → Monitor and evaluate → Reflect and learn

B. Choose → advertise → implement → stop

C. Analyze → implement → define → reflect

D. Monitor → define → select → ignore

Correct answer: A
Why? The slide lists these seven stages in this order.
Source: PDF p. 28 — Slide 31

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	B	C	C	B	B	B	A	A
Q	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	A	B	B	A	A	B	A	A
Q	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	A	A	A	A	A	A	A	B
Q	31	32	33	34	35	36	37	38	39	40
Ans.	A	A	A	A	A	B	C	B	B	A
Q	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	A	B	A	A	A	A	B	A